MATH 195: 11/12
TRIG IDENTITIES

An *identity* is an equation that's always true, no matter the value of the input. Identities are algebraically useful because if one half of the equation appears in a larger formula you can replace it with the other half without changing the formula.

Identities we've already seen.

- The pythagorean identity.

$$\cos^2 t + \sin^2 t = 1$$

- Even-odd identities.

$$\cos(-t) = \cos t \qquad \sin(-t) = -\sin t \qquad \tan(-t) = -\tan t$$

- Definitions of other trig functions.

$$\sec t = \tfrac{1}{\cos t} \quad \csc t = \tfrac{1}{\sin t} \quad \cot t = \tfrac{1}{\tan t}$$

$$\tan t = \tfrac{\sin t}{\cos t} \quad \cot t = \tfrac{\cos t}{\sin t}$$

Other forms of the pythagorean identity.

By dividing both sides by either $\cos^2 t$ or $\sin^2 t$ you get versions of the pythagorean identity about the other trig functions.

$$1 + \tan^2 t = \sec^2 t \qquad \cot^2 t + 1 = \csc^2 t$$

Angle-sum and double-angle identities.

Other identities let you rewrite what happens for complicated inputs in terms of simpler inputs.

- $\sin(x + y) = \sin x \cos y + \cos x \sin y$
- $\cos(x + y) = \cos x \cos y - \sin x \sin x$
- $\sin(2x) = 2 \sin x \cos x$
- $\cos(2x) = \cos^2 x - \sin^2 x$

Exercises, Part 1

(1) Use the pythagorean identities to come up with formulas for the value of $\mathsf{trig}^2 t$ for all six trig functions. [Hint: You already have formulas for $\sec^2 t$ and $\csc^2 t$. For the others you need a single algebra step each.]

(2) Explain how the double-angle identities come from the angle-sum identities by applying the angle-sum identities to $\sin(x + x)$ and $\cos(x + x)$.

(3) Use the angle-sum and even-odd identities to come up with *angle-difference* identities for $\sin(x - y)$ and $\cos(x - y)$.

(4) Determine $\sin(7\pi/12)$ by using an angle-sum identity. [Hint: $\sin(\pi/4) = \cos(\pi/4) = \sqrt{2}/2$, $\sin(\pi/3) = \sqrt{3}/2$, and $\cos(\pi/3) = 1/2$.]

(5) Determine $\cos(\pi/12)$ by using an angle-difference identity. [Hint: same as before.]

(6) Use the pythagorean identity to explain why

$$\cos(2x) = 2\cos^2 x - 1 \qquad \text{and} \qquad \cos(2x) = 1 - 2\sin^2 x.$$

(7) Come up with an angle-sum identity for $\tan(x + y)$ by rewriting it as $\frac{\sin(x+y)}{\cos(x+y)}$ and then using the identities for sine and cosine.

Half-angle identities.
You can go the other direction too.

$$\cos(x/2) = \pm\sqrt{\frac{1+\cos x}{2}} \qquad \sin(x/2) = \pm\sqrt{\frac{1-\cos x}{2}},$$

Where you can determine sign by looking at which quadrant the angle $x/2$ is in.

Exercises, Part 2

(1) Use a half-angle identity to determine $\sin(\pi/12)$ and $\cos(\pi/12)$. [Hint: $\sin(\pi/6) = 1/2$ and $\cos(\pi/6) = \sqrt{3}/2$.]
(2) Use a half-angle identity to determine $\cos(\pi/8)$. [Hint: $\cos(\pi/4) = \sqrt{2}/2$.]
(3) Now determine $\cos(\pi/16)$ and $\cos(\pi/32)$.
(4) Determine a general pattern for $\cos(\pi/2^n)$.
(5) Explain where the half-angle identities come from by using the versions of the double-angle identity for cosine from the earlier exercises. [Hint: When looking at the equation

$$\cos(2x) = 2\cos^2 x - 1$$

think of x as $t/2$ and $2x = t$, then solve for $\cos(t/2)$.]

Exercises, Part 3

In calculus, the *difference quotient* of a function $f(x)$ is used as part of defining the *derivative* of $f(x)$. Namely, the difference quotient is

$$\frac{f(x+\Delta x) - f(x)}{\Delta x}.$$

Here, Δx ("Delta x") is a single variable representing a small change in x. Some textbooks will use another variable, e.g. h, here, but in my view the meaning is most clear when you use Δx.

A difference quotient gives an approximation to the slope of a tangent line by stepping forward a small distance Δx and looking at the slope of the line through the points $(x, f(x))$ and $(x+\Delta x, f(x+\Delta x))$.

(1) Draw a picture to confirm the difference quotient's relationship to slope.
(2) Use the angle sum identity to rewrite the difference quotient for $\sin x$ in a simple a form as possible. Your goal is to (a) make the inputs to trig functions as simple and possible and (b) to separate as much as possible x and Δx.
(3) Think about the difference quotient at $x = 0$ for $\sin x$. What happens as Δx gets infinitely small—closer and closer to 0. What value does it look like the difference quotient is approaching? Use computer tools to help with your calculations.
(4) Use the angle sum identity to rewrite the difference quotient for $\cos x$ in a simple a form as possible.
(5) Think about the difference quotient at $x = 0$ for $\cos x$. What happens as Δx gets infinitely small—closer and closer to 0. What value does it look like the difference quotient is approaching? Use computer tools to help with your calculation.
(6) Earlier problems should hinge, respectively, the expressions

$$\frac{\sin(\Delta x)}{\Delta x} \qquad \text{and} \qquad \frac{\cos(\Delta x) - 1}{\Delta x}.$$

[If these expresses didn't appear in your solutions for (2) and (4), go back and simplify further.] Use computer tools to help determine what values these expressions approach as Δx approaches 0.
(7) Based on this, what values do the difference quotients for $\sin x$ and $\cos x$, at an arbitrary input x, approach as Δx approaches 0.